

LAB 1 - USE-CASE FLOW

Student / SRN

Project

Course / Lab

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Hospital Bed & ICU Allocation Dashboard

Software Engineering - Lab 1

UC-02 - Allocate Bed / ICU

Primary actor: Triage Nurse | Supporting actor: Hospital EHR / Patient Records System | Goal: Allocate a suitable available hospital bed or ICU bed to a patient.

Preconditions

1. Triage nurse is authenticated.
2. Patient information is available in the hospital records system.
3. Current bed/ICU availability is accessible.

Main Success Scenario

4. The triage nurse selects Allocate Bed / ICU.
5. The system requests the patient's identification/details.
6. The system retrieves relevant patient information from the Hospital EHR / Patient Records System.
7. The system displays suitable available beds and ICU beds.
8. The triage nurse selects a suitable available bed or ICU bed.
9. The system verifies that the selected bed is still available.
10. The system records the allocation against the patient.
11. The system changes the bed status from Available to Occupied/Reserved.
12. The system updates the dashboard with the new allocation.
13. The system displays an allocation confirmation; the use case ends successfully.

Alternate Flow - Requested Bed Unavailable

- 6A. The system detects that the selected bed is no longer available.
- 6A1. The system displays an unavailable message and lists suitable alternatives.
- 6A2. The triage nurse selects an alternative bed; the system verifies its availability.
- 6A3. If the alternative is available, resume at Step 7. If no suitable alternative is available, the system informs the nurse and ends without allocation.

Postconditions

- Success: Selected bed is allocated, its status is updated, and the dashboard reflects the change.
- Failure: No incorrect allocation is recorded and the triage nurse is informed.